

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING & ENGINEERING TECHNOLOGY
DEPARTMENT OF CIVIL ENGINEERING
SECOND SEMESTER 2018/2019 SESSION EXAMINATIONS
CVE 202 – STRENGTH OF MATERIALS I

Instructions: Answer 4 questions in all.

Time allowed: 2.5 hrs.

1. Answer Question 1 (compulsory) and three other Questions.

2. Attempt at least 1 Question from Section A, 1 Question from Section B, and 1 Question from Section C

QUESTION 1 (COMPULSORY)

[25 Marks]

(a) A cantilever beam AB with length L has a concentrated load W at the free end B. Derive the equations for the slope and deflection at B using double integration method. **[7 marks]**

(b) A steel rod 10 m long and 3 cm in diameter is subjected to an axial tensile load of 50 kN. Determine the change in length, diameter and volume of the rod. Take Young's Modulus = $2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio = 0.25 **[8 marks]**

(c) Draw the free body diagrams of each of the following structures shown in FIG. 1.

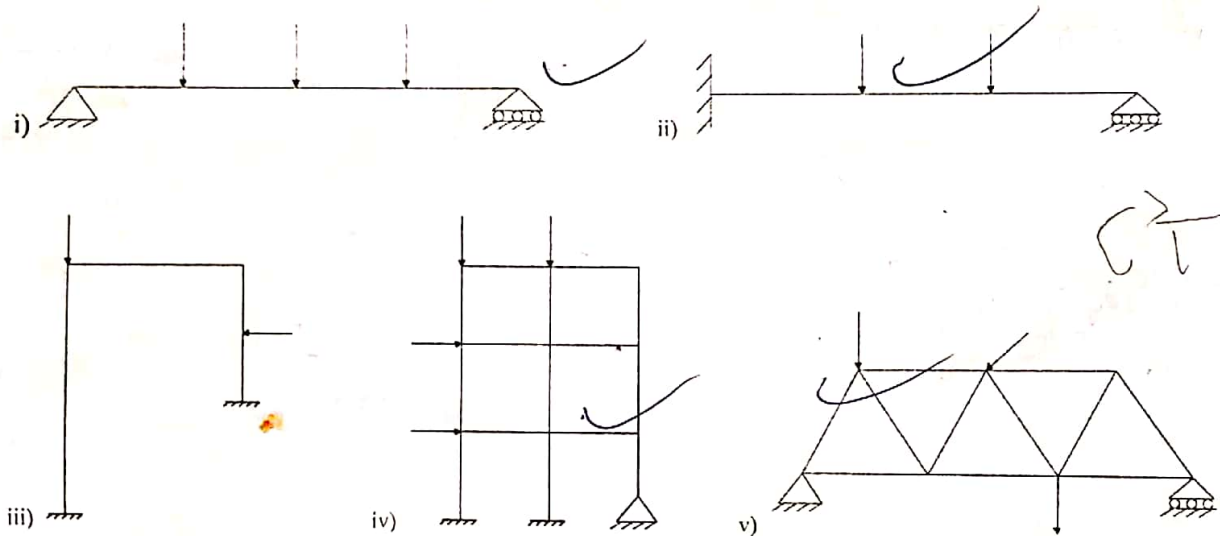
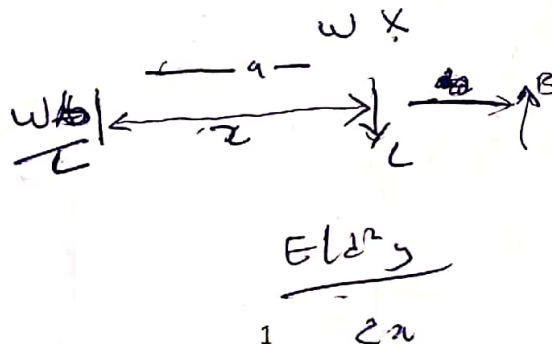


FIG. 1

[10 marks]



1 2π

$$\frac{100}{25 \times 4} = 100$$

$$\frac{300}{2} = 150$$

SECTION A

QUESTION 2

[25 Marks]

An overhanging beam ABC supports a concentrated load P at the end of the overhang (FIG. 2). Span AB has length L and the overhang has length a . Determine the deflection δ_C at the end of the overhang by using the following methods:

(a) Obtain the strain energy of the beam and use Castigliano's theorem. [13 marks]

(b) Use the modified form of Castigliano's theorem (differentiation under the integral sign). [12 marks]

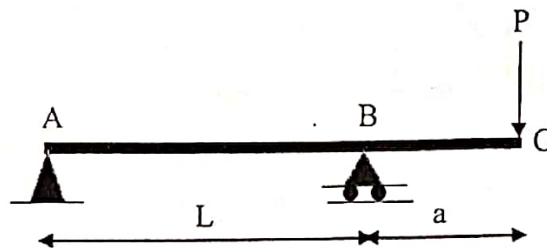


FIG. 2

QUESTION 3

[25 Marks]

A beam AB of length L simply supported at the ends carries a point load W at a distance a from the left end as shown in FIG. 3.

(a) Using Macaulay's method, derive the equation(s) for the:

(i) Deflection under the load [10 marks]

(ii) Position of maximum deflection [5 marks]

(b) If $W = 40 \text{ kN}$, $a = 4 \text{ m}$, $L = 6 \text{ m}$, $I_{xx} = 73.3 \times 10^{-6} \text{ m}^4$ and $E = 200 \text{ GN/m}^2$, compute the numerical values of the deflection under the load and the position of the maximum deflection. [10 marks]

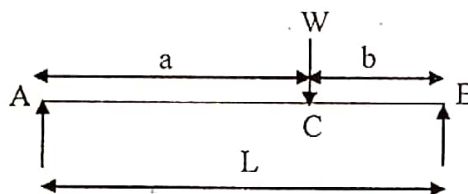


FIG. 3

$$\frac{d^2y}{dx^2}$$

SECTION B

QUESTION 4

[25 Marks]

(a) Prove that the volumetric strain of a cylindrical rod subjected to an axial tensile load is equal to strain in length minus twice the strain in diameter. [15 marks]

(b) Determine the value of Young's modulus and Poisson's ratio of a metallic bar of length 1 m, breadth 40 mm and depth 40 mm when the bar is subjected to an axial pull of 400 kN. The increase in length is given as 0.125 cm and increase in breadth is given as 0.0015 cm.

[10 marks]

QUESTION 5

[25 Marks]

(a) A steel rail is 12 m long and is laid at a temperature of 18 °C. The maximum temperature expected is 38 °C. If the stress developed is 20 N/mm², what is the gap provided between the rails. Take Young's Modulus = 2×10^5 N/mm²; Coefficient of linear expansion = 12×10^{-6} /°C.

[8 marks]

(b) The stepped bar shown in FIG. 4 is made up of two different materials. The material 1 has a Young's modulus = 2×10^5 N/mm², while material 2 has a Young's modulus = 1×10^5 N/mm². Find the total extension of the bar under a pull of 70 kN if both portions have a thickness of 30 mm.

[9 marks]

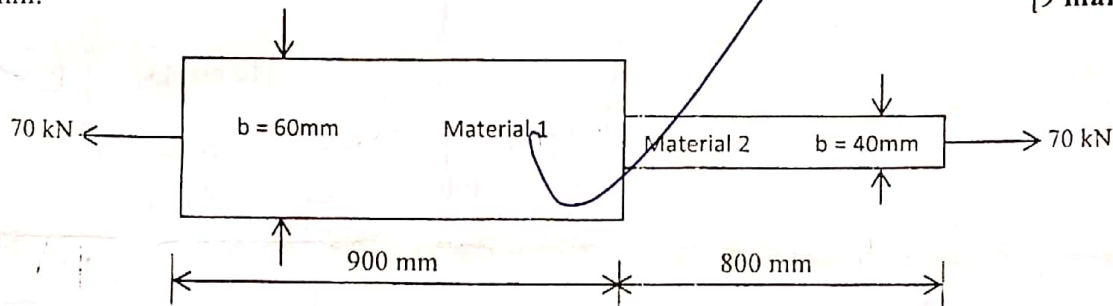


FIG. 4

(c) Determine the required thickness for the wall of a thin cylinder of internal diameter 1.8 m, if the permissible stress is 60 N/mm² and internal fluid pressure of 3 N/mm².

[8 marks]

SECTION C

QUESTION 6

[25 Marks]

For the beam shown in FIG. 5 below,

- Calculate the support reactions,
- Draw the shear force and bending moment diagrams,
- Determine the point of contraflexure,
- Find the maximum bending moment.

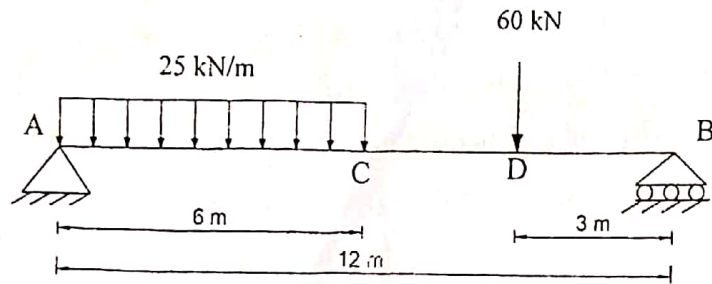


FIG. 5

[25 marks]

✓ **QUESTION 7**

[25 Marks]

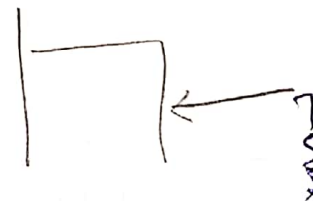
- (a) It is required to transmit 100 kW by means of a solid circular shaft rotating at 3 rev/s. The allowable shearing stress is 45 MPa. Find the required shaft diameter.

[10 marks]

- (b) A hollow steel shaft 5m long must transmit a torque of 30 kNm. The total angle of twist in this length is not to exceed 2.5° and allowable shearing stress is 100 MPa. Determine the inside and outside diameter of the shaft if G is 85 GPa.

[15 marks]

1
6
4



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CIVIL ENGINEERING DEPARTMENT
STRENGTH OF MATERIALS CVE 202

Instructions:

1. Answer 4 questions in all

2. Section A is compulsory. Answer one question each from other sections

Time allowed: 2.5 hrs.

SECTION A

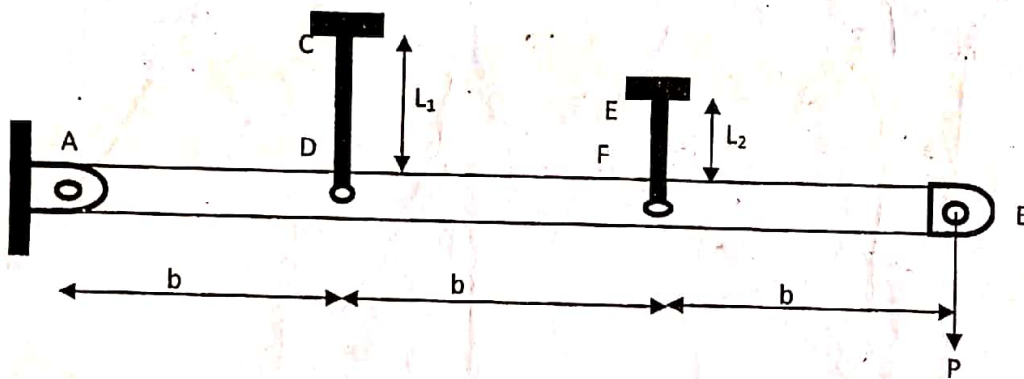
Q1. (a) Outline expressions for stiffness and flexibility of prismatic member. Define all variables

✓(b) State Castiglianos' theorem?

✓(c) Write an equation for the maximum shear stress induced in a cylindrical shell.

SECTION B

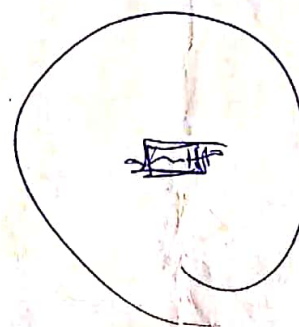
Q2. A horizontal rigid bar AB is pinned at end A and supported by two wires (CD and EF) at points D and F (FIG Q2A). A vertical load P acts at end B of the bar. The bar has length $3b$ and wires CD and EF have lengths L_1 and L_2 , respectively. Also, wire CD has diameter d_1 and modulus of elasticity E_1 ; wire EF has diameter d_2 and modulus E_2 . Obtain formulas for the allowable load P if the allowable stresses in wires CD and EF , respectively, are σ_1 and σ_2 . (Disregard the weight of the bar itself.)

**FIG Q2A**

✓ Q3(a) A simple beam AB supports a uniform load of intensity q and a concentrated load P (FIG Q3). The load P acts at the midpoint C of the beam. The beam has length L , modulus of

1

1, 3, 5, 6



elasticity E and moment of inertia I . Determine the downward deflection δ_c at the midpoint of the beam by the following methods:

- ✓(i) Obtain the strain energy of the beam and use Castigliano's theorem, and
- ✓(ii) Use the modified form of Castigliano's theorem (differentiation under the integral sign).

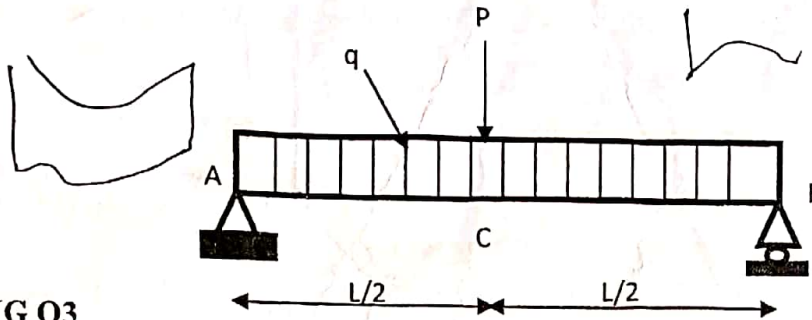


FIG Q3

(b) A cylinder of internal diameter 0.50 m contains air at a pressure of 7 N/mm^2 (gauge). If the maximum permissible stress induced in the material is 80 N/mm^2 . Find the thickness of the cylinder.

SECTION C

Q4(a) Explain the behaviour of metal subjected to external load

(b) Find the Tension in the ropes T_1 and T_2 shown in **FIG Q4A** if the weight of the suspended body is 500N.

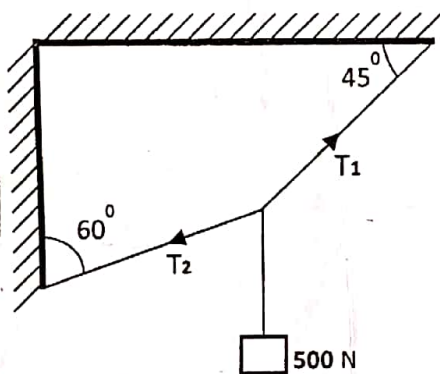


FIG Q4A

Q5FIG Q6B shows an overhanging beam.

- (i) Draw the Shear force and the Bending moment diagrams
- (ii) Calculate the value of the maximum bending moment and
- (iii) The distance of the point of contraflexure from supports A and B

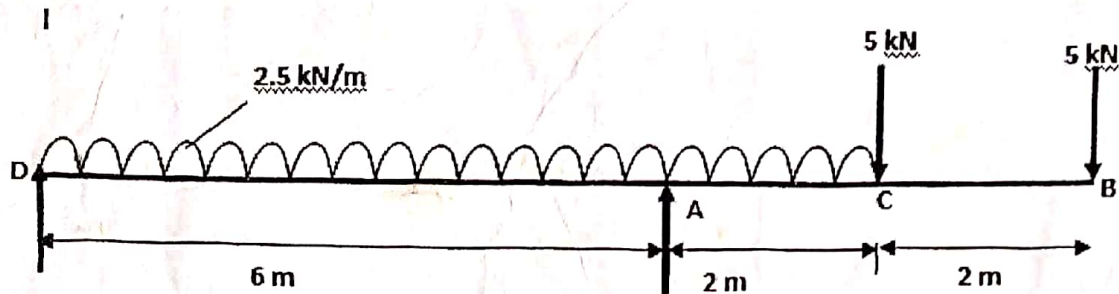
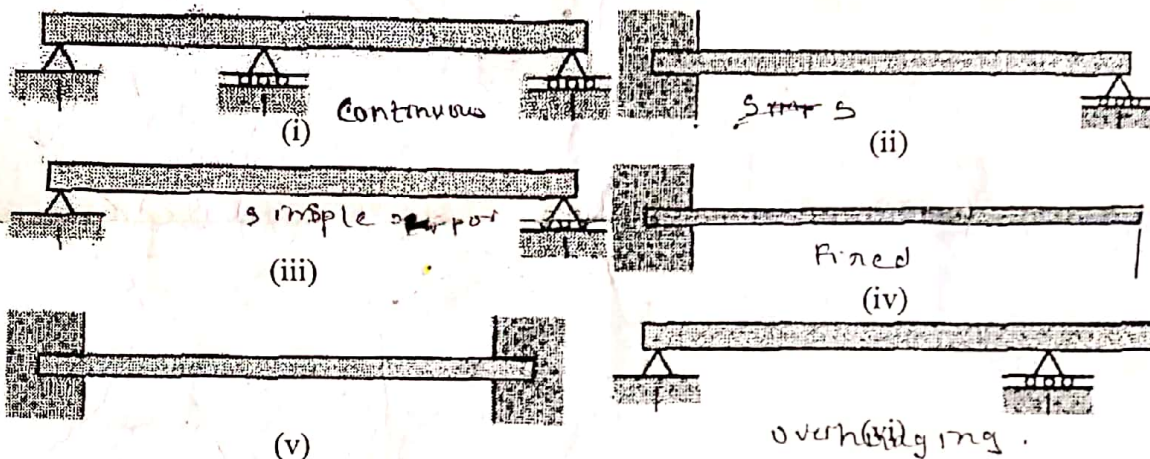


FIG Q5B

SECTION D

Q6(a) Identify the class of the following beams



(b) For portion of A and B of the overhanging beam showing in Figure 1.

- (i) Derive the equation for the elastic curve using double integration method
- (ii) Draw the graph of y against x at 1 m time step.

$$P = 50 \text{ kN} \quad I = 723 \text{ mm}^4 \quad E = 29 \times 10^6 \text{ N/mm}^2$$

$$L = 15 \text{ m} \quad a = 4 \text{ m}$$

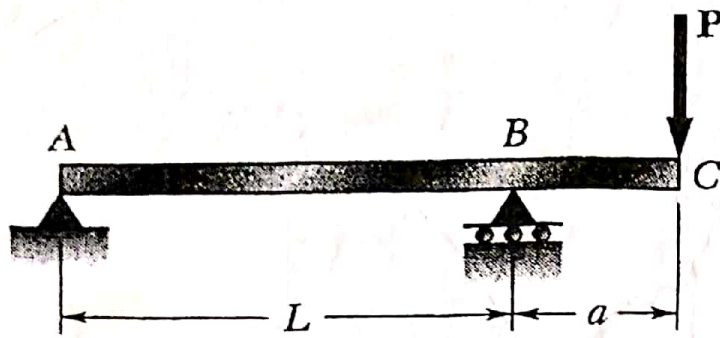


FIG Q6B

Q7 (a) Determine the deflection of the beam at 2 m using Macaulay's method. $E = 208 \text{ GN/m}^2$, $I = 82 \times 10^{-6} \text{ m}^4$

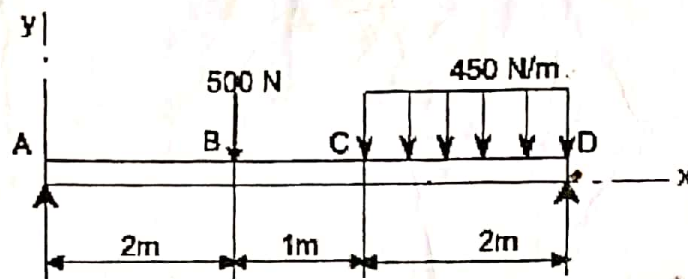


FIG Q7A

(b) Draw the graph of y against x at 1 m time step.

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2018/2019 ACADEMIC SESSION
CVE 202 TEST

Time Allowed: 1hr

1. For the beam shown in Fig. 1 below,
 - a) calculate the support reactions,
 - b) draw the shear force and bending moment diagrams,
 - c) determine the point of contraflexure,
 - d) find the maximum bending moment

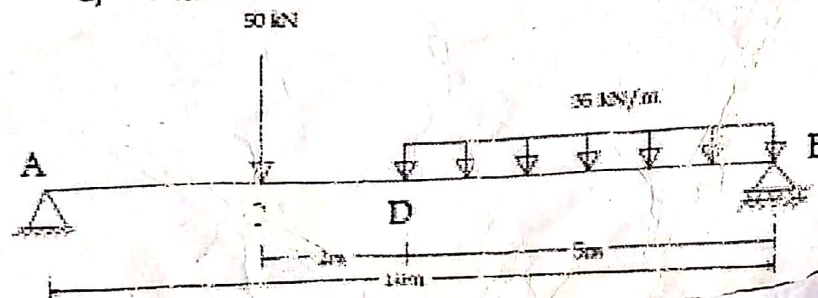


Fig. 1

2. A beam AB of length L simply supported at the ends carries a point load W at a distance a from the left end as shown in Fig. 2. Derive the equation for the deflection under the load using Macaulay's method. Hence, if $W = 40 \text{ kN}$, $a = 4 \text{ m}$, $L = 6 \text{ m}$, $I_x = 73.3 \times 10^{-6} \text{ m}^4$ and $E = 200 \text{ GN/m}^2$, calculate the magnitude of the deflection under the load.

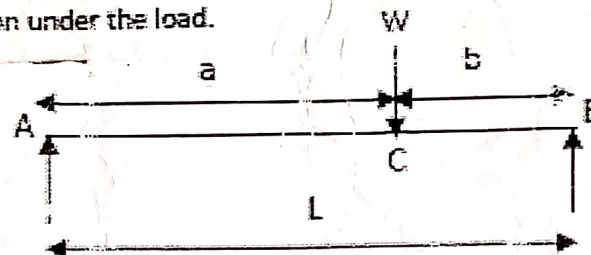


Fig. 2

3. A Steel rail is 12 m long and is laid at a temperature of 20°C . The maximum temperature expected is 38°C . Calculate the temperature stress developed in the rails if:
 - a) No expansion joint is provided.
 - b) If a 1.2 mm gap is provided for expansion. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\alpha = 12 \times 10^{-6}/^\circ\text{C}$.